

Machine Learning Models Evaluation Results

Final Results Summary (Sorted by Accuracy)

Algorithm	Accuracy	Weighted F1	Macro F1	Train Time	Predict Time
MLP — Filter_Chi2	0.7833	0.7598	0.5661	9.11s	0.02s
XGBoost (Baseline) — Wrapper_RFECV	0.7779	0.7476	0.5409	0.47s	0.01s
XGBoost (Baseline) — Full_Features	0.7774	0.7473	0.5429	0.78s	0.03s
XGBoost (Optimized) — Wrapper_RFECV	0.7756	0.7401	0.5276	135.57s	0.03s
SVM (Optimized) — Wrapper_RFECV	0.7752	0.7522	0.5504	174.86s	1.55s
MLP — Wrapper_RFECV	0.7731	0.7301	0.5301	5.64s	0.02s
MLP (Optimized) — Filter_Chi2	0.7724	0.7471	0.5635	70.73s	0.02s
XGBoost (Optimized) — Wrapper_Forward	0.7708	0.7300	0.5150	93.34s	0.02s
XGBoost (Baseline) — Wrapper_Forward	0.7677	0.7267	0.5124	0.31s	0.01s
XGBoost (Optimized) — Filter_MI	0.7667	0.7238	0.5070	114.42s	0.02s
MLP — Full_Features	0.7641	0.7153	0.5002	4.06s	0.03s
XGBoost (Baseline) — Filter_MI	0.7614	0.7120	0.4868	0.40s	0.02s
MLP (Optimized) — Wrapper_RFECV	0.7602	0.7210	0.5249	77.59s	0.01s
SVM (RBF) — Full_Features	0.7529	0.7187	0.5195	1.81s	2.27s
MLP — Filter_MI	0.7517	0.7029	0.4799	6.61s	0.03s
SVM (Optimized)	0.7483	0.7260	0.5391	140.95s	2.18s

— Filter_Chi2					
SVM (RBF) — Filter_Chi2	0.7463	0.7193	0.5300	0.94s	1.82s
XGBoost (Optimized) — Filter_Chi2	0.7458	0.7181	0.5152	117.93s	0.06s
SVM (RBF) — Wrapper_RFECV	0.7433	0.7107	0.5078	0.89s	1.65s
XGBoost (Baseline) — Filter_Chi2	0.7430	0.7138	0.5097	0.41s	0.02s
SVM (Linear) — Wrapper_RFECV	0.7417	0.6966	0.4777	0.18s	0.00s
SVM (Linear) — Full_Features	0.7399	0.6961	0.4843	0.24s	0.01s
MLP — Wrapper_Forward	0.7293	0.6962	0.4889	5.43s	0.02s
MLP (Optimized) — Filter_MI	0.7291	0.6819	0.4697	67.63s	0.02s
SVM (Optimized) — Filter_MI	0.7281	0.6983	0.4982	155.96s	1.74s
MLP (Optimized) — Wrapper_Forward	0.7280	0.6944	0.4853	58.72s	0.02s
SVM (RBF) — Filter_MI	0.7247	0.6780	0.4730	0.93s	1.88s
SVM (Linear) — Wrapper_Forward	0.7237	0.6730	0.4659	0.11s	0.00s
SVM (Optimized) — Wrapper_Forward	0.7237	0.7263	0.5337	201.48s	4.83s
SVM (Linear) — Filter_MI	0.7207	0.6728	0.4559	0.21s	0.00s
SVM (Linear) — Filter_Chi2	0.7171	0.6704	0.4691	0.15s	0.00s
SVM (RBF) — Wrapper_Forward	0.7064	0.6585	0.4439	1.57s	2.23s

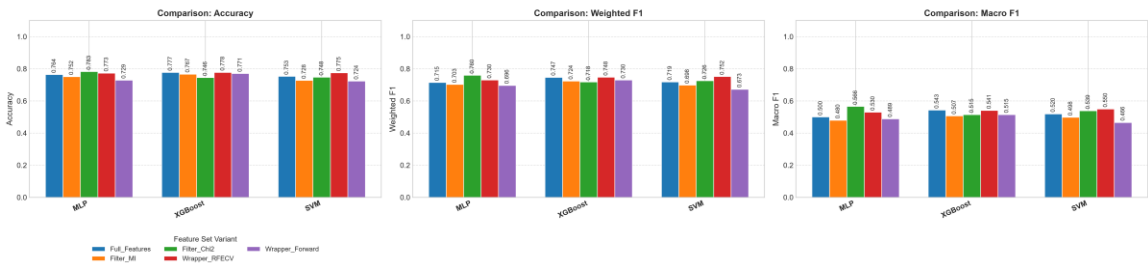
Comprehensive Model Overview

Comprehensive Model Overview: Performance & Computational Cost



Algorithm vs. Feature Selection Comparison

Algorithm vs. Feature Selection Comparison

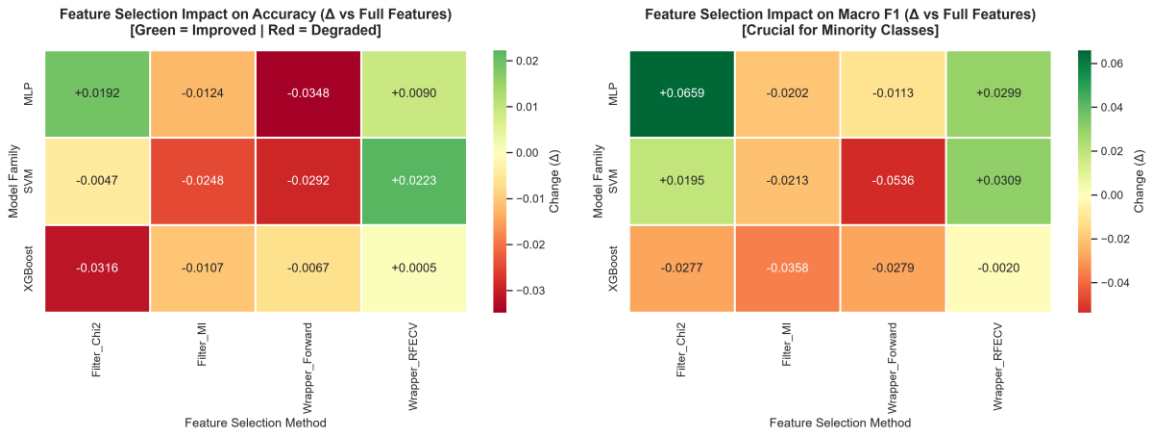


Filter Impact vs. Full-Features Baseline

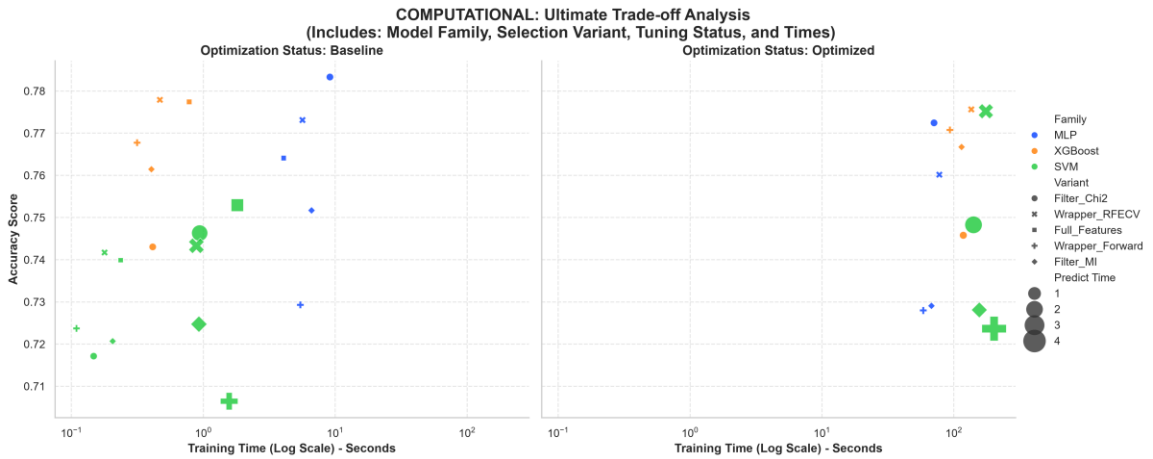
Model Family	Feature Set	Δ Accuracy	Δ Macro F1	Train Time	Predict Time
MLP	Filter_MI	-0.0124	-0.0202	6.61s	0.0261s
MLP	Filter_Chi2	+0.0192	+0.0659	9.11s	0.0247s
MLP	Wrapper_RFECV	+0.0090	+0.0299	5.64s	0.0237s
MLP	Wrapper_Forward	-0.0348	-0.0113	5.43s	0.0221s

XGBoost	Filter_MI	-0.0107	-0.0358	114.42s	0.0231s
XGBoost	Filter_Chi2	-0.0316	-0.0277	117.93s	0.0616s
XGBoost	Wrapper_RFECV	+0.0005	-0.0020	0.47s	0.0130s
XGBoost	Wrapper_Forward	-0.0067	-0.0279	93.34s	0.0158s
SVM	Filter_MI	-0.0248	-0.0213	155.96s	1.7421s
SVM	Filter_Chi2	-0.0047	+0.0195	140.95s	2.1755s
SVM	Wrapper_RFECV	+0.0223	+0.0309	174.86s	1.5470s
SVM	Wrapper_Forward	-0.0292	-0.0536	0.11s	0.0026s

Feature Selection Impact Heatmaps

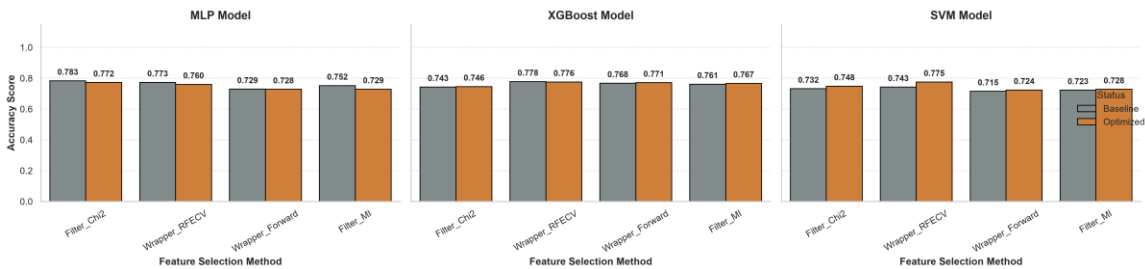


Computational vs Accuracy Trade-off Analysis



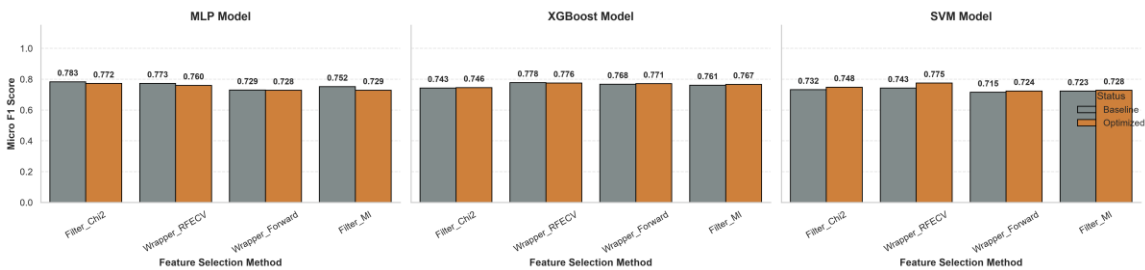
GridSearchCV Impact on Accuracy

Impact of GridSearchCV on [Accuracy] across Feature Selection Methods



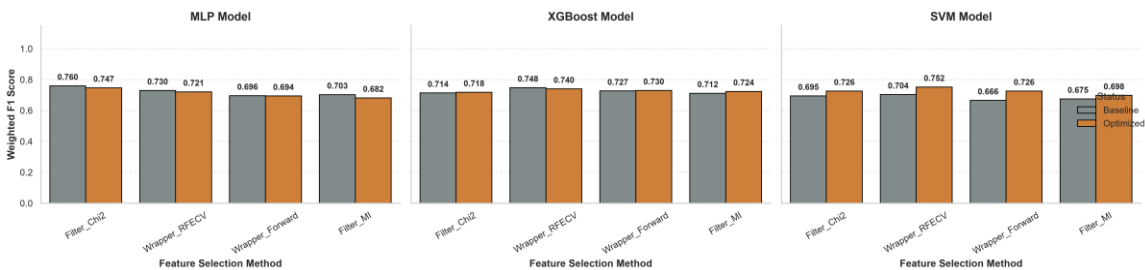
GridSearchCV Impact on Micro F1

Impact of GridSearchCV on [Micro F1] across Feature Selection Methods



GridSearchCV Impact on Weighted F1

Impact of GridSearchCV on [Weighted F1] across Feature Selection Methods



Overall Impact of Hyperparameter Tuning

